



AIR UNIVERSITY ISLAMABAD

Department of Mechatronics Engineering

**SOLID MODELLING AND MANUFACTURE
PROCESS**

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Section	A
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PROJECT TITLE	UNIVERESAL JOINT
Submitted To: (Instructor)	Muhammad Irfan Minhas
Section Details	The two main functional sections of a universal joint are the Yokes (connecting the shafts) and the Spider or Cross (the central pivot linking them).
Project Status (e.g., Final)	The universal joint is a mature, indispensable mechanical component whose current status focuses on increasing durability, reducing weight through advanced materials (like composites), and integrating sensors for predictive maintenance.

Abstract: Abstract of Universal Joint Assembly is given below

Abstract

The objective of this Comprehensive Engineering Project (CEP) was to design, model, and document a complete cross-type Universal Joint (U-joint) assembly to effectively transmit power between misaligned rotating shafts. The modeling approach involved a bottom-up methodology in SolidWorks, meticulously creating 3D models of the individual parts—yokes, spider/cross, bearings, and retaining rings—using core features like Extrude and Revolve. The components were then integrated using advanced Assembly Mates to accurately simulate the rotational and pivoting motion required for angular transmission. The final output consists of the fully constrained, functioning 3D assembly model, rendered images, and a comprehensive set of 2D engineering drawings for each part, complete with dimensions, tolerances, and sectional views, thereby fully documenting the design for manufacturing and verification.

Indroduction of Project

Universal joint: A universal joint (also called a universal coupling or U-joint) is a joint or coupling connecting rigid shafts whose axes are inclined to each other. It is commonly used in shafts that transmit rotary motion. It consists of a pair of hinges located close together, oriented at 90° to each other, connected by a cross shaft. The universal joint is not a constant-velocity joint.



Cardan joint, after Gerolamo Cardano, a polymath of the 16th century who contributed to knowledge of various clever mechanisms, including gimbals

Hooke joint or Hooke's joint, after Robert Hooke, a polymath of the 17th century who contributed to knowledge of various clever mechanisms

Spicer joint, after Clarence W. Spicer and the Spicer Manufacturing Company, who manufactured U joints

Hardy Spicer joint, after the Hardy Spicer brand, a successor to the Spicer brand

Project Part Summary (Universal Joint Components)

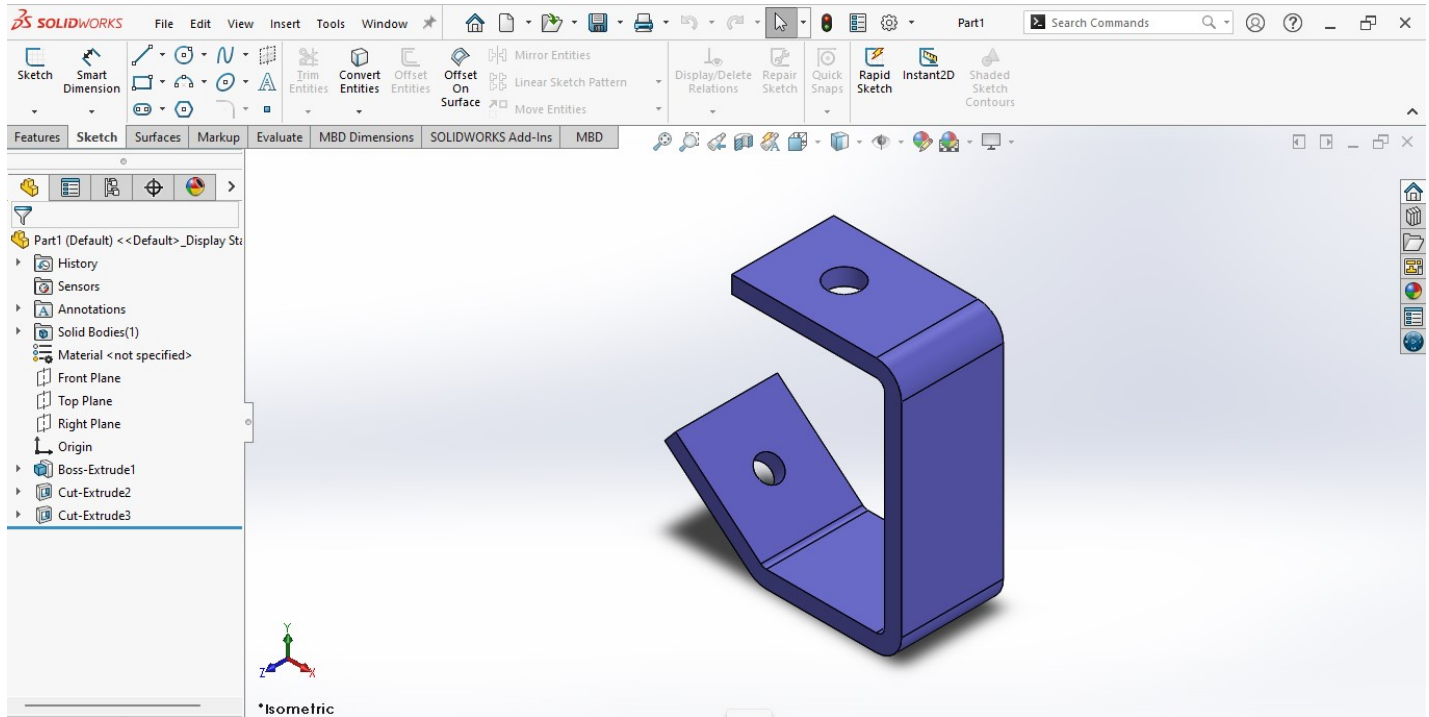
The project involved modeling various components, which are listed below with their descriptions and quantities. Note that the part names in the tables from the document seem to refer to a different mechanism (Crankshaft/Drive Disk, Connecting Rod/Link Arm) than the main Universal Joint components (Yokes, Spider/Cross) described elsewhere in the document's abstract. The table below uses the names and descriptions provided in the document's tables, followed by the abstract's reference to the main universal joint components.

Item No.	Part Name	Description (Function)	QTY
1	Base Plate / Frame	The main stationary structure for mounting the assembly.	1
2	Input Shaft Yoke / Crank	The component connected to the driving shaft, initiating rotational motion.	1

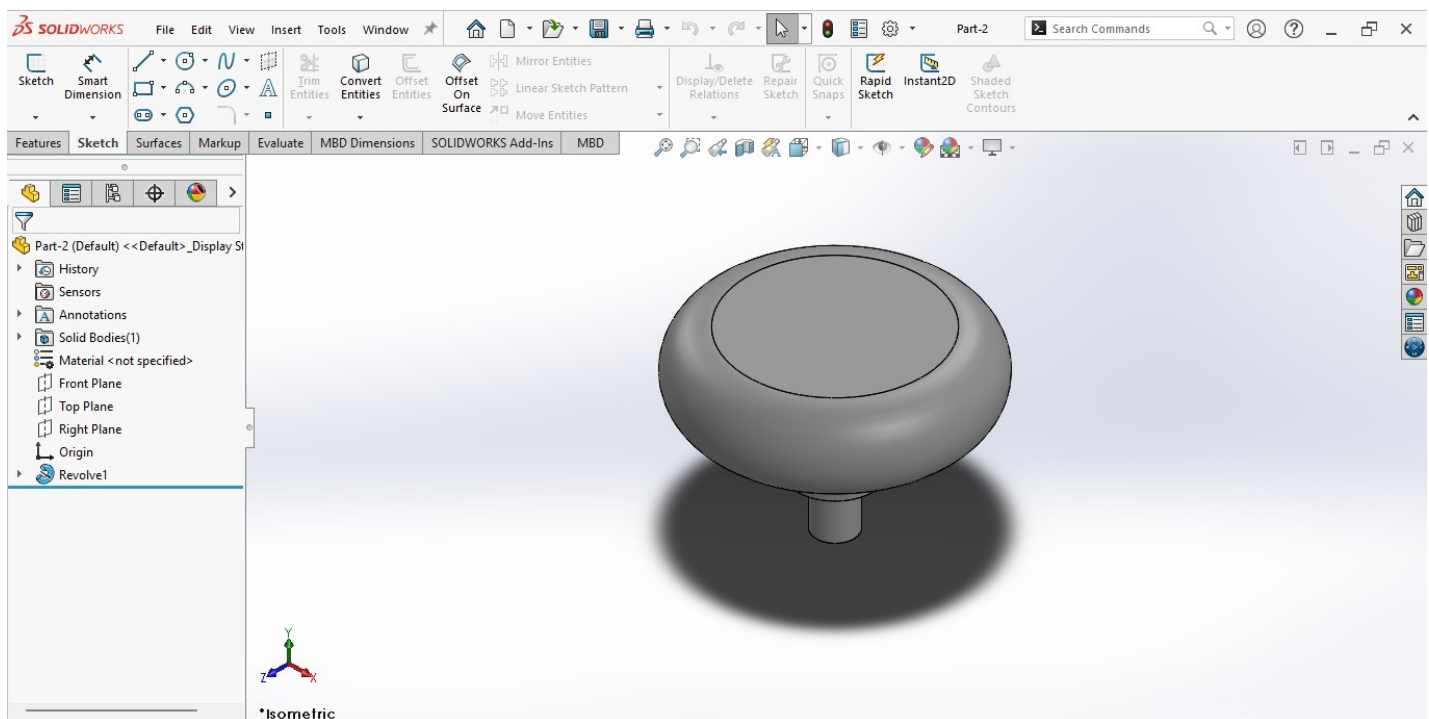
Item No.	Part Name	Description (Function)	QTY
3	Spider / Cross Pin	The central pivot element, which connects the two yokes and allows angular misalignment.	1
4	Output Shaft Yoke / Link Arm	The component connected to the driven shaft, receiving the transmitted motion.	2
5	Bearing Cap / Bushing	A component used to retain the spider's pin within the yoke for smooth articulation.	1
6	Shaft Bushing / Bearing Sleeve	An internal bearing element to reduce friction on a rotating shaft.	1
7	Fastener 1	Generic securing component (e.g., screw or bolt).	1
8	Fastener 2	Generic securing component (e.g., set screw or bolt).	1
9	Retaining Clip / Fastener 3	Generic securing component (e.g., retaining ring or washer).	1

Procedure PART:1

Sketch an arc of the required dimensions.

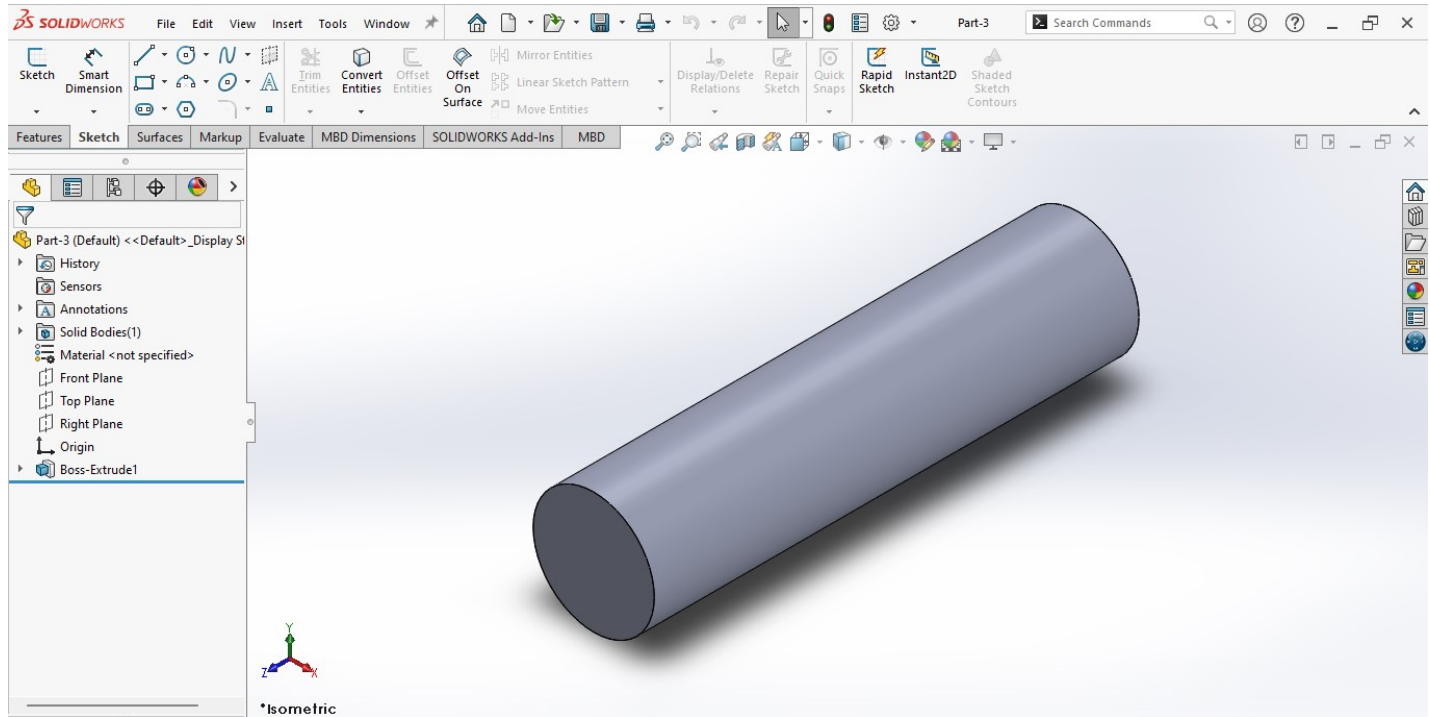


PART:2

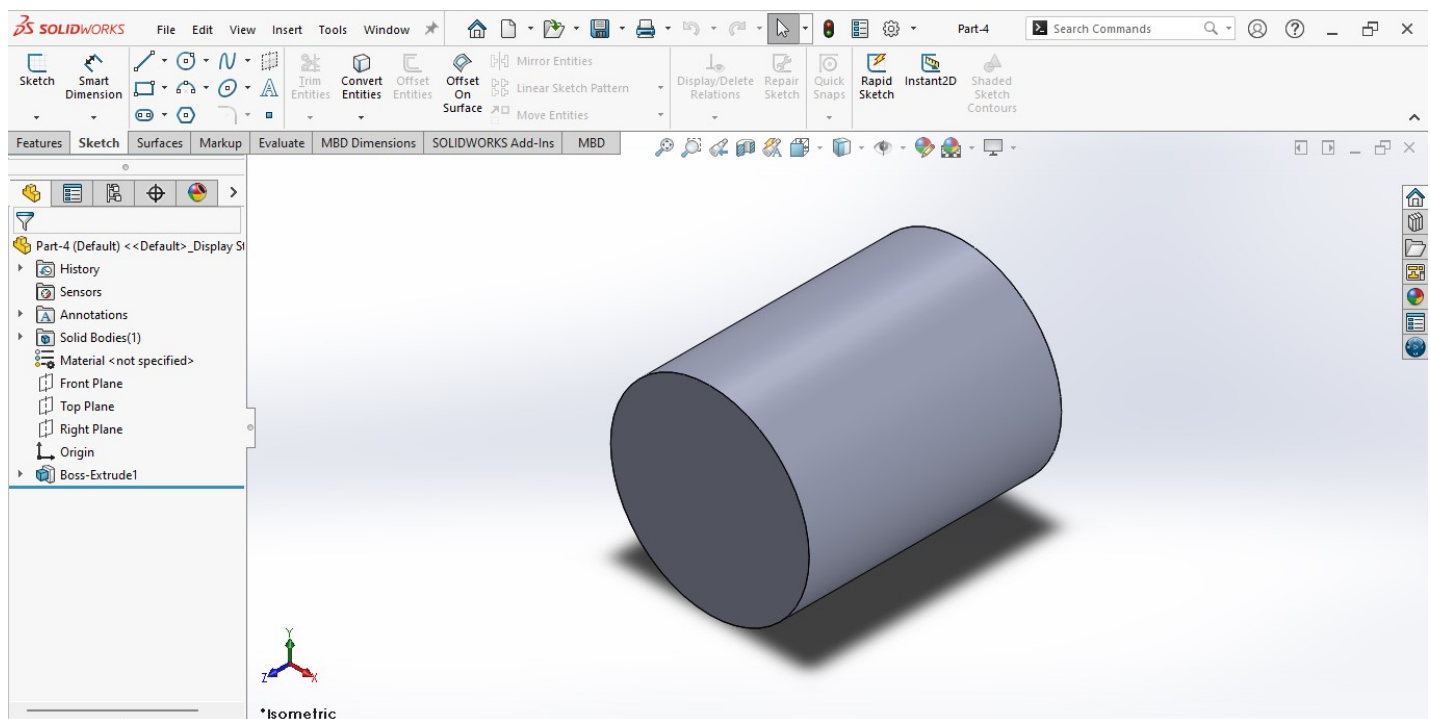


Procedure PART:3

Sketch an arc of the required dimensions.

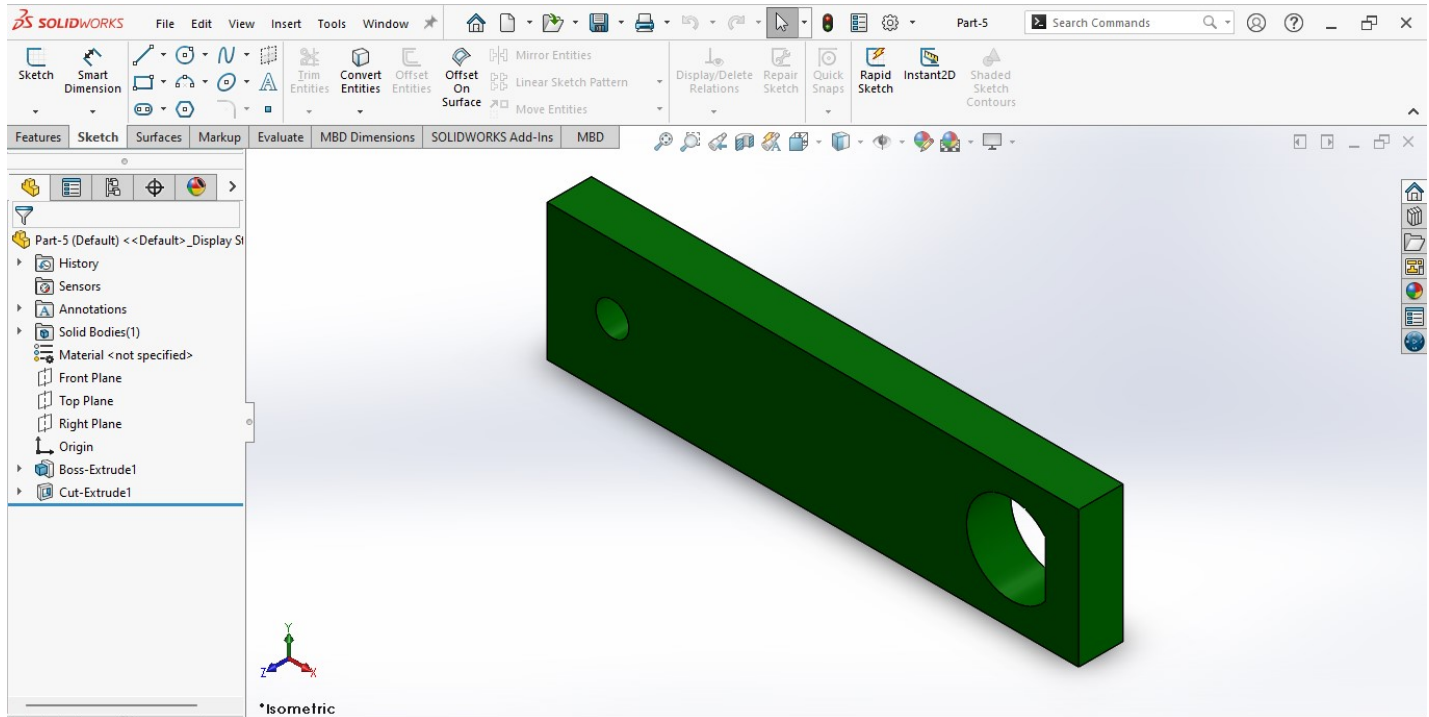


PART:4

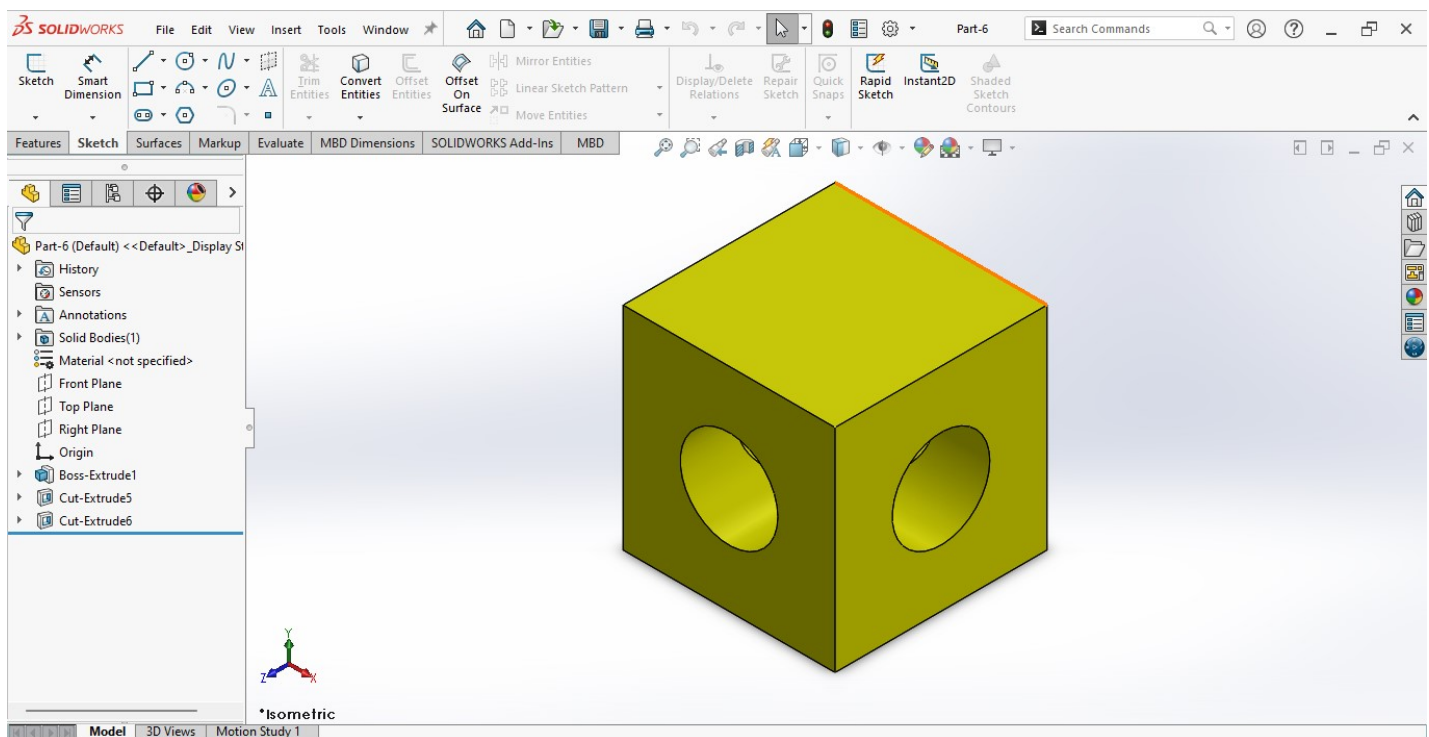


Procedure PART:5

Sketch an arc of the required dimensions.

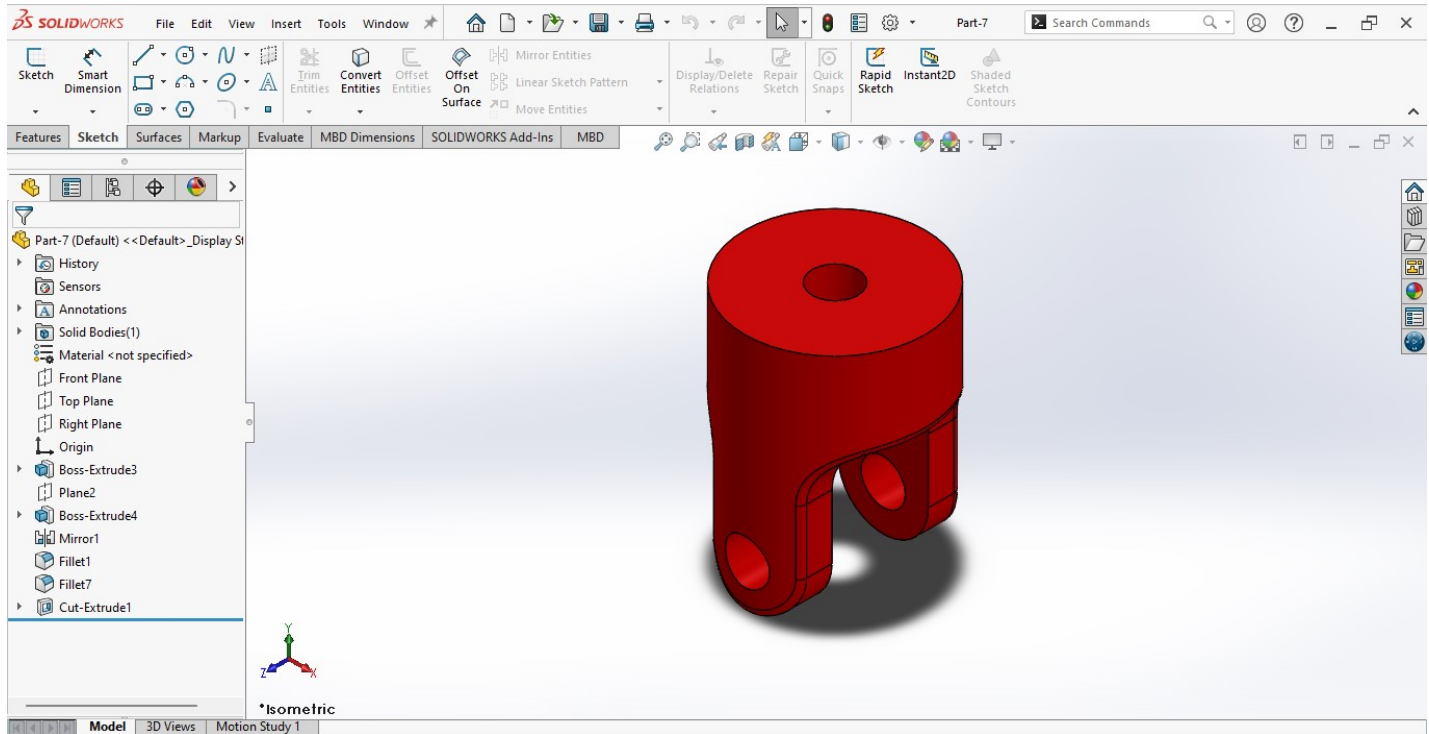


PART:6

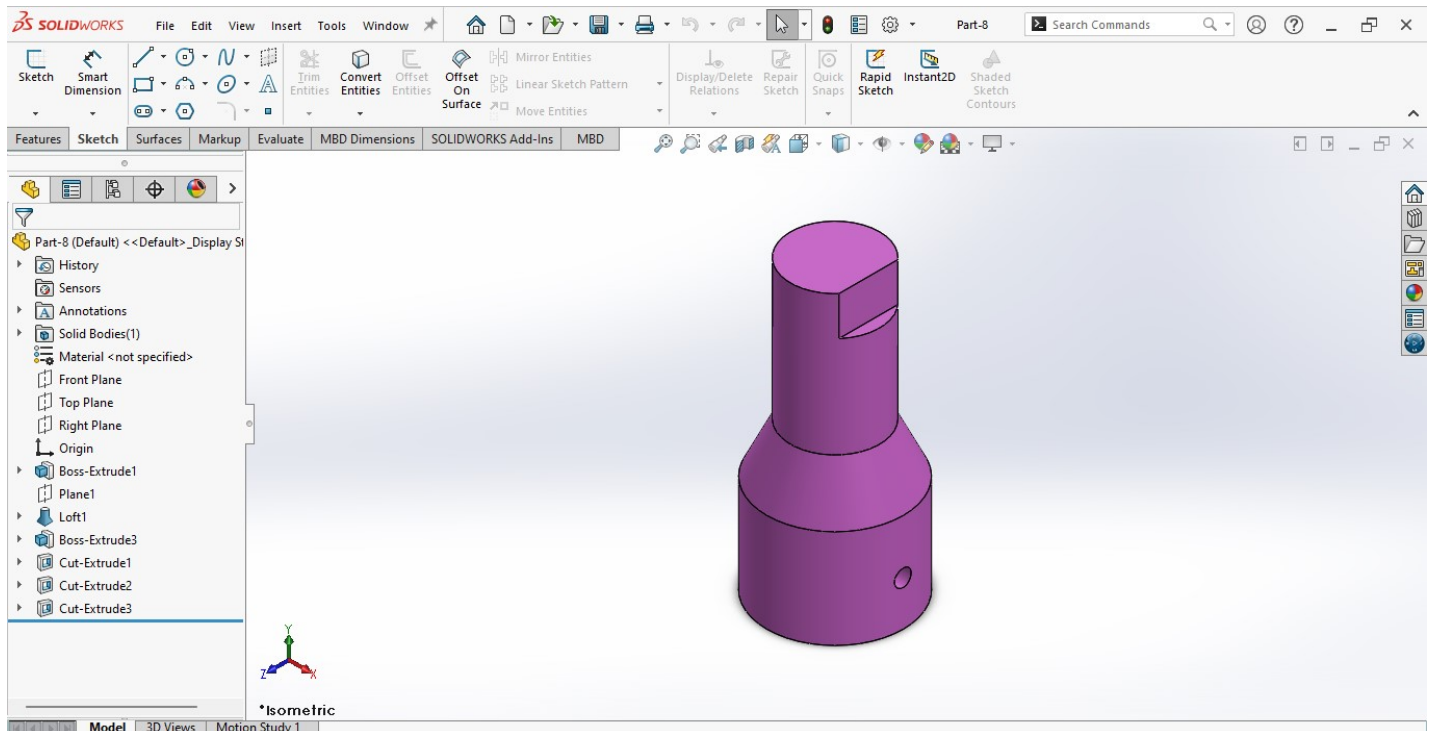


Procedure PART:7

Sketch an arc of the required dimensions.

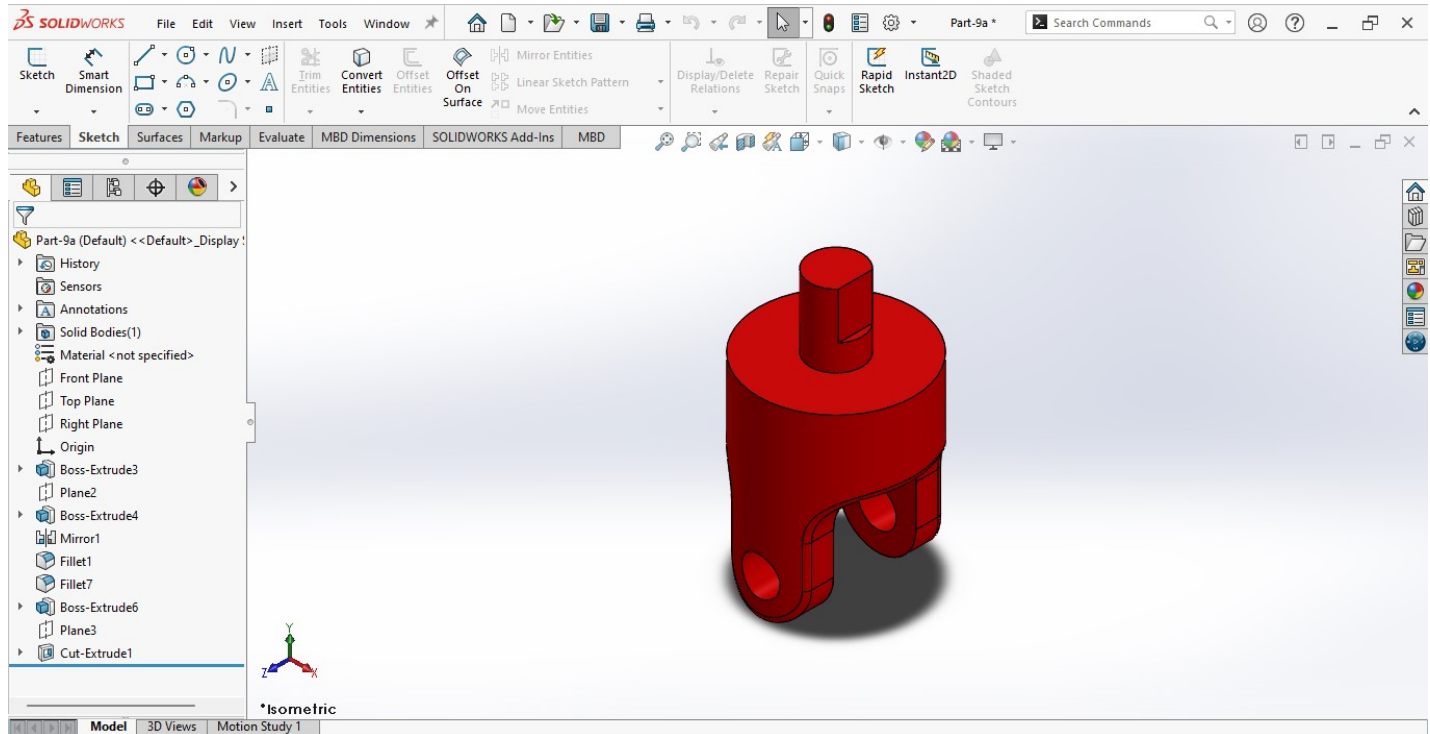


PART:8

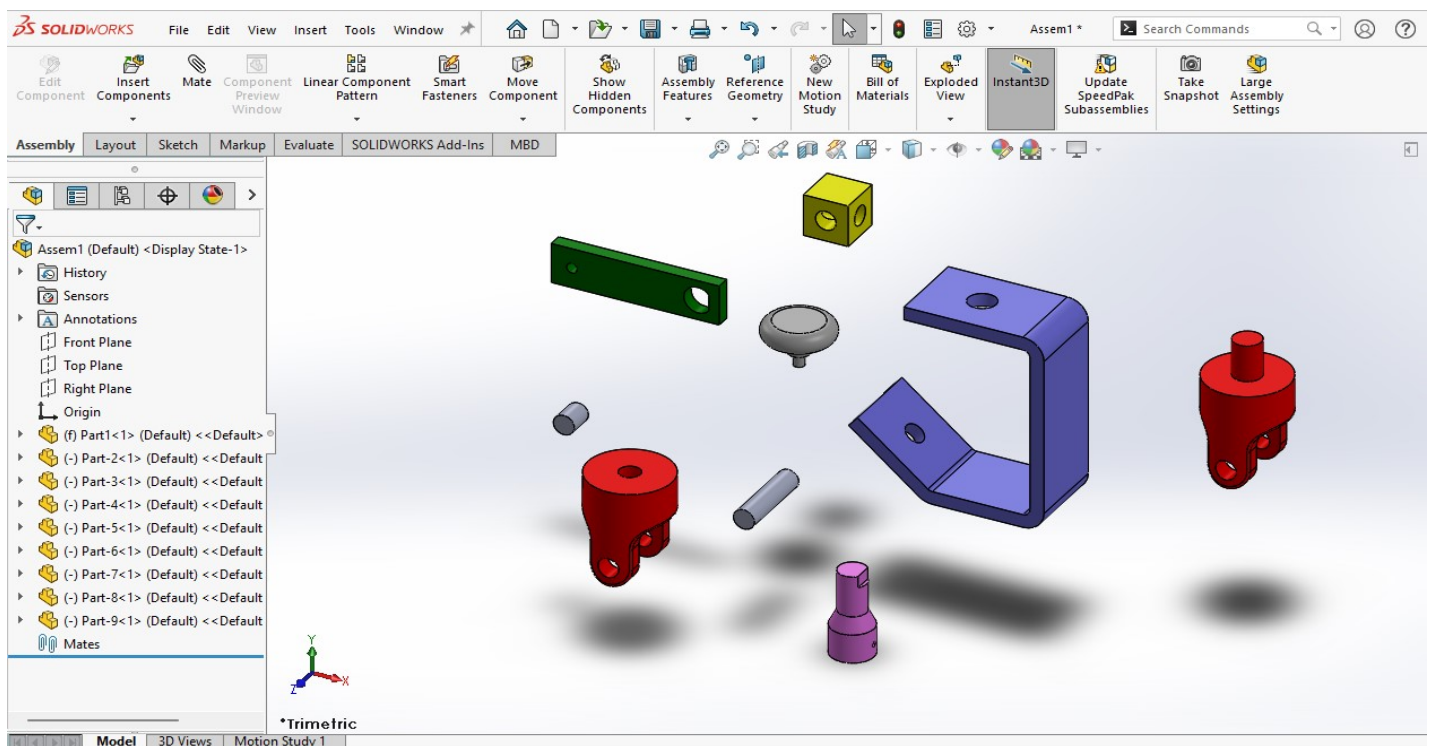


Procedure PART:9

Sketch an arc of the required dimensions.

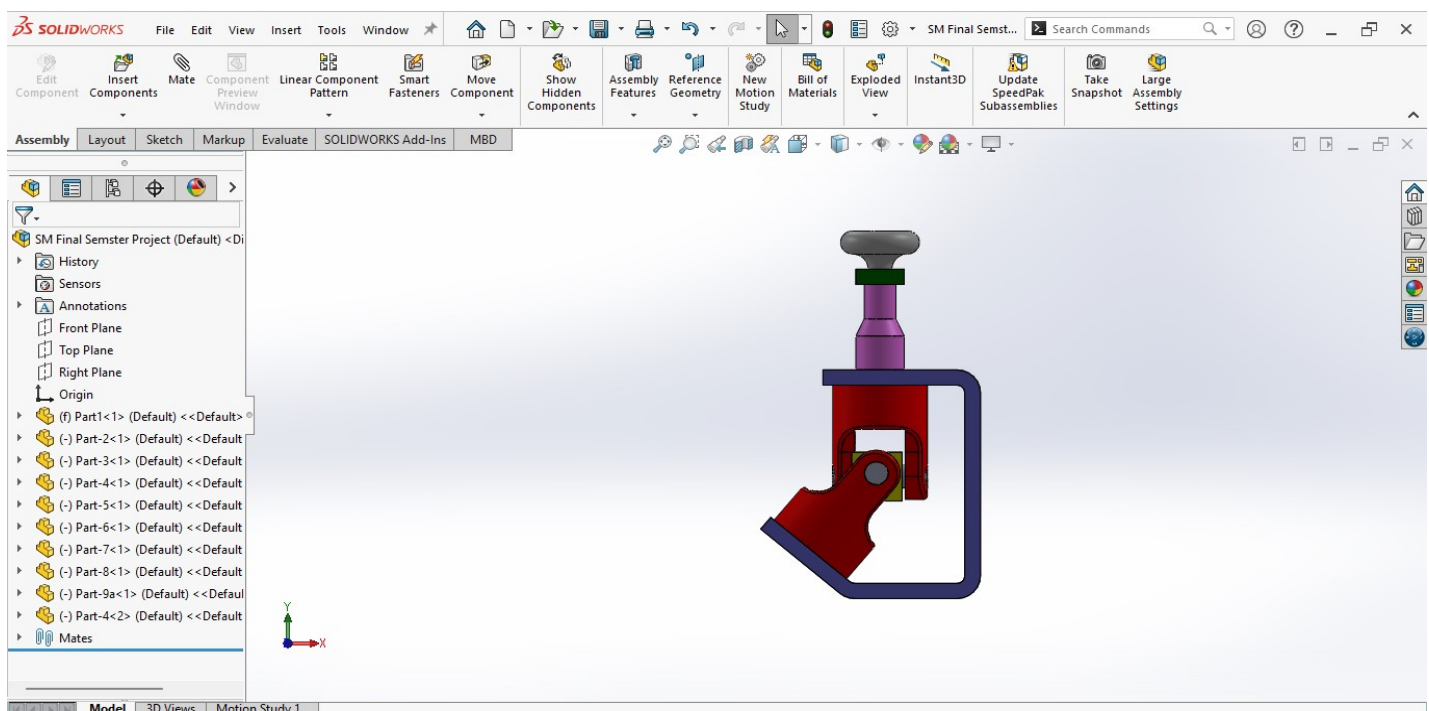


FINALLY ALL PARTS:

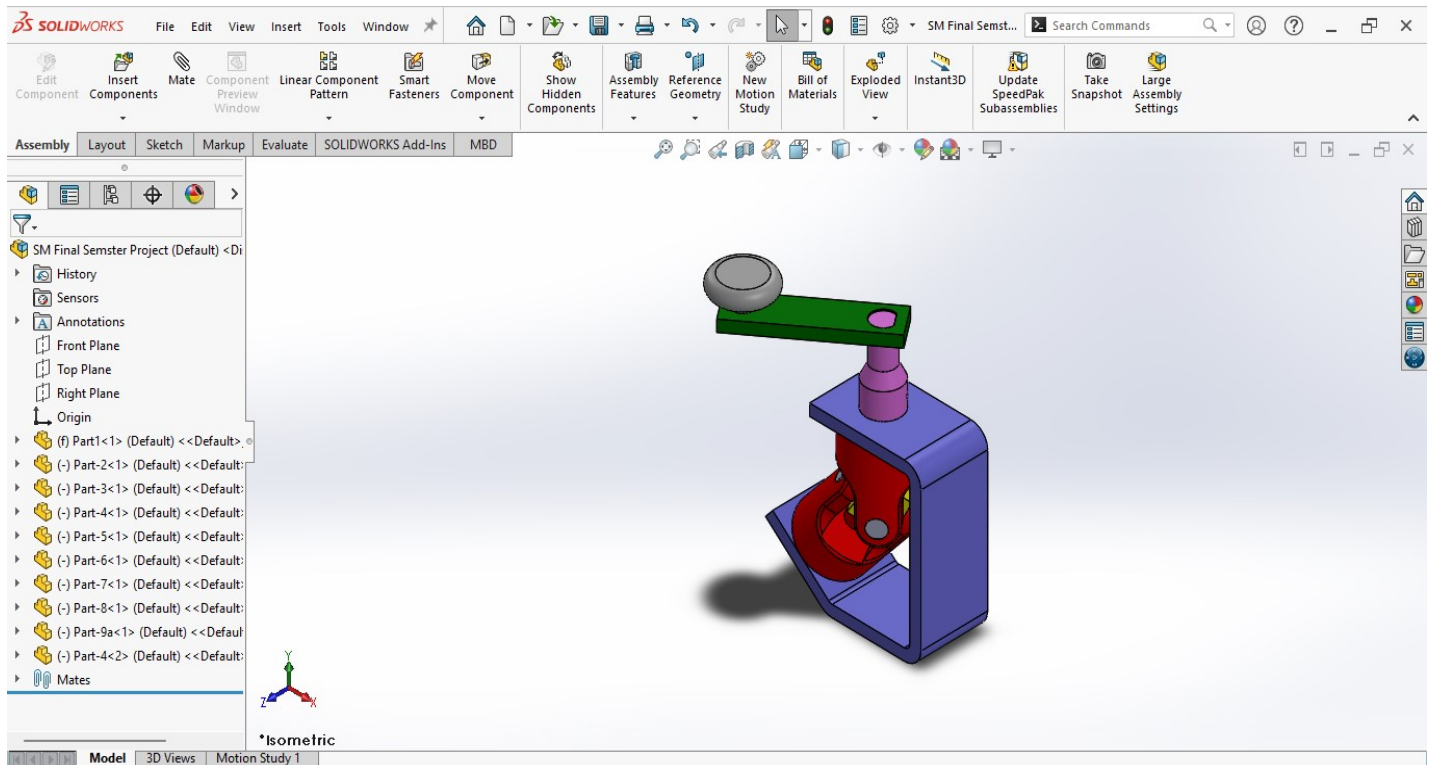


Project Title	Universal Joint (11)
Assembly Method	Bottom-up methodology using SolidWorks12.
Key Components (U-Joint)	Yokes (connecting the shafts) and the Spider or Cross (the central pivot linking them)13.
Mating	Integrated using advanced Assembly Mates to accurately simulate the rotational and pivoting motion14.
Final Output	The fully constrained, functioning 3D assembly model, rendered images, and a comprehensive set of 2D engineering drawings for each part15.

NORMAL VIEW:



ISOMETRIC VIEW:



✓ Conclusion

The objective of this Comprehensive Engineering Project (CEP) was successfully achieved by designing, modeling, and documenting a complete cross-type **Universal Joint (U-joint)** assembly in SolidWorks.

Modeling Proficiency: The project demonstrated mastery of the bottom-up modeling methodology by successfully creating and assembling all necessary components, including the Yokes and the Spider/Cross, using core SolidWorks features.

Functional Validation: The final assembly model successfully validates the U-joint's fundamental function: efficiently transmitting rotary motion between two shafts whose axes are inclined to each other. This confirms the design can handle angular misalignment, which is critical for its application in drivetrains and control mechanisms.

Documentation: The project produced a complete technical package, including 2D engineering drawings with dimensions and tolerances, thereby fulfilling the requirements for the Solid Modeling and Manufacture Process (SM & MP) course by preparing the design for production readiness.

THE END